

Night and Day: Diurnal Warming and Structural Transformation in India

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Abstract

This paper finds diverging partial effects of diurnal warming—higher nighttime and daytime temperatures—on agricultural wage-labour shares from decadal Indian Censuses (1981–2011). Though both margins contract grain output and cultivated area, only higher maxima raise harvest prices locally—consistent with a model where warmer nights shock land but warmer days shock land and labour productivity. Warmer nights shift seasonal workers and self-cultivators into agricultural labour; warmer days push labour to the seasonal margin. Long differences show the labour divergence is rural. In towns, both margins depress non-agricultural worker shares.

Keywords: Climate Change, Structural Transformation, Agriculture, India

JEL Codes: O1, Q54, J43, R11

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Structural transformation—the reallocation of labour from agriculture into non-agricultural employment—is widely documented as the foundation of economic growth and development (Lewis 1954; Matsuyama 1992; Kongsamut, Rebelo, and Xie 2001; Herrendorf, Rogerson, and Valentinyi 2014). A growing literature studies how climate change impacts structural transformation in developing economies (Saxena 2024; Dasgupta and Raghav 2024; Pham 2026). In particular, rising temperatures have been shown to have greater impacts than rainfall fluctuations on agricultural productivity (Schlenker and Roberts 2009) and labour reallocation (Colmer 2021; Liu, Shamdasani, and Taraz 2023). Yet, the evidence on whether warming temperatures raise or lower agricultural labour shares remains fractured.

On the one hand, Jesso, Manning, and Taylor (2017) find extreme heat in rural Mexico depresses formal wage employment and pushes out-migration. In India, though heat shocks increase temporary out-migration (Das and Sanyal 2026), caste-based insurance networks outweigh wage gains from permanent migration (Munshi and Rosenzweig 2016). Within districts, Colmer (2021) finds that temperature shocks *reduce* demand for agricultural labour, increasing nonagricultural employment in the short run.

On the other hand, Liu, Shamdasani, and Taraz (2023) document that higher temperatures *raise* district-level agricultural labour shares, through reduced nonagricultural demand in the long run. Across countries, Nath (2025) shows that despite heat-induced productivity losses, a “food problem” intensifies agricultural specialisation in developing economies, exacerbating the “agricultural productivity gap”.¹ The debate remains unsettled on whether sectoral reallocation of labour can mitigate local economic losses under a warming climate.

This paper offers one resolution to the debate on warming temperatures and structural transformation: whether agricultural employment shares rise or fall follows the relative magnitudes of rising nighttime minima (T_{min}) or daytime maxima (T_{max})—compressing or expanding the diurnal temperature range (DTR). The headline result is a divergence in agricultural wage-labour shares based on the *time of day* where warming is most prominent: warmer nights raise, warmer days lower this share.

This night-day divergence in agricultural labour is rooted in crop and human physiology: crops stand on land around the clock, while farmers are exposed most under daylight. Consequently, warmer nights raise nocturnal respiration, spending the carbon crops fix by day (Peng et al. 2004; Welch et al. 2010); warmer days press field labour against a physiological ceiling, lowering the productivity of labour itself (Dunne, Stouffer, and John 2013; Somanathan et al. 2021). Agronomy has documented that nighttime and daytime warming carry distinct effects on crop yields and pull in different directions on overall production (Peng et al. 2004; Lobell 2007; Welch et al. 2010; Chen, Chen, and

¹Agriculture continues to employ roughly 42% of India’s contemporary workforce (World Bank, 2025).

Xu 2016). The diurnal decomposition, however, has not crossed into the economic literature on structural transformation. From a decadal panel and long-differenced approach within Indian Census districts and towns (1981–2011), I find that higher nighttime (T_{min}) temperatures *hold* agricultural labour on land while higher daytime (T_{max}) temperatures *release* it—even though both measures reduce agricultural production and land-use.

Prior economic literature has primarily measured warming through ‘degree days’, a count of extreme days, or binned daily temperatures (Schlenker, Hanemann, and C. 2006; Blakeslee and Fishman 2015; Dell, Jones, and Olken 2014; Somanathan et al. 2021). These daily measures average temperatures over the diurnal cycle, and therefore collapse the night-day distinction that agronomy holds apart. The closest neighbours to this paper—studying Indian labour reallocation—likewise treat warming as a scalar: Colmer (2021) and Liu, Shamdasani, and Taraz (2023) identify structural-transformation effects averaging diurnal temperatures. By separating the nighttime and daytime margins, this paper uncovers an additional dimension to the relationship between temperature and agricultural productivity shocks through factor-bias (Bustos, Caprettini, and Ponticelli 2016; Barrett, Ortiz-Bobea, and Pham 2023) under asymmetric diurnal warming. An equivalent representation is the level and the diurnal range of surface-air temperatures. Together, they separate two distinct mechanisms of structural transformation in the local economy.

Figure 1 shows the asymmetric warming geography of India, obtained by long-differencing the 10-year average of nighttime and daytime temperatures in districts between two endpoints (1981,91 – 2001,11). Although night and day temperature ‘levels’ rose across the board, the bias on the DTR sorts across space. Asymmetric diurnal warming (Karl et al. 1991, 1993; Easterling et al. 1997; Davy et al. 2016; IPCC 2021) reached India after 1991 (Mall et al. 2021): the DTR contracted in the arid northwest and Indo-Gangetic plains, and expanded in the Deccan peninsula and along the coasts. This geography provides continuous variation in nighttime versus daytime warming across districts and towns.

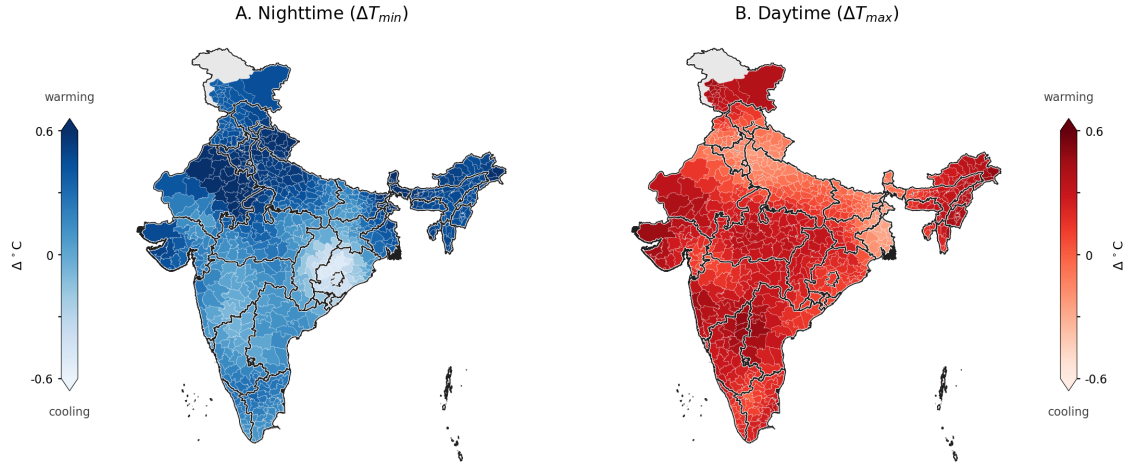
This paper establishes three core empirical facts.

First, within districts, decadal panel estimates (Table 1) show that higher nighttime and daytime temperatures both reduce production and area under cultivation of grains, while only daytime temperatures raise their harvest prices locally.² Under both shocks, cultivated area falls with a similar magnitude ($\beta_{T_{min}} = \beta_{T_{max}} \approx -0.22$ log points, $F = 0.01$). Notably, nighttime warming produces a nominally milder loss in grain *production* (-0.15) relative to daytime warming (-0.22), with no detectable change in local harvest prices. Daytime warming acts as a physical supply shock, spiking local harvest prices by 0.13 log points, rejecting equality with night warming ($F = 13.34$).

²The core grain basket of India is composed of rice, wheat, maize, barley, sorghum and millets.

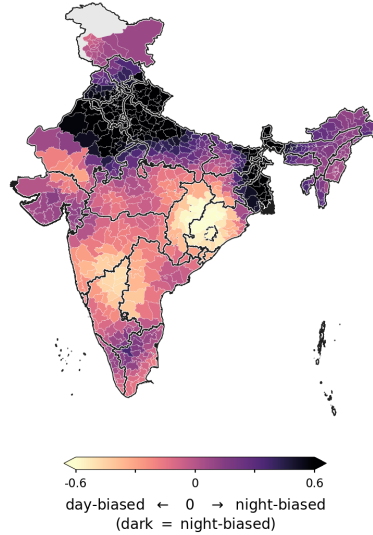
FIGURE 1. The diurnal warming geography of India

Long-differenced change in growing-season temperature, 1981 → 2011



Diurnal warming bias, 1981 → 2011

C. Nighttime – Daytime ($\Delta T_{min} - \Delta T_{max}$)



Notes: Spatial distribution of warming nighttime and daytime temperatures in India. This map plots the change in decadal temperatures over the growing-season (June–Feb), by long-differencing the average between two endpoints: (2001,2011) – (1981,1991). IMD gridded climate is assigned to each district centroid by inverse-distance weighting. Panels A and B map the change in the nighttime minimum (ΔT_{min}) and the change in daytime maximum (ΔT_{max})—darker where warming is stronger. Panel C maps the diurnal warming bias ($\Delta T_{min} - \Delta T_{max}$): darker where nights warmed faster (compressing DTR) and lighter where days warmed faster (widening DTR). Across the 639 districts with map coverage, nights outpaced days across the Indo-Gangetic north, while days outpaced nights across the peninsular Deccan (the estimation panel covers roughly 300 districts, Section 2).

Second, the direction of change in agricultural wage-labour shares reverses between night and day warming. A 1°C decadal rise in nighttime minimum temperatures ($\beta_{T_{\min}}$) *increases* agricultural wage-labour shares by +4.4 percentage points within districts, drawing from self-cultivators (−2.8) and seasonal ‘marginal’ workers (−3.3).³ An equivalent 1°C decadal rise in the daytime maximum temperatures ($\beta_{T_{\max}}$) *decreases* agricultural wage-labour shares (−2.6) and raises seasonal worker shares (+3.2). The two coefficients reject joint equality ($\beta_{T_{\min}} = \beta_{T_{\max}}$) on agricultural labour shares ($F = 24.9$) and seasonal worker shares ($F = 14.7$).

The opposing signs on agricultural labour shares are robust to humidity-adjusted wet-bulb temperatures (+0.086/ − 0.079) and a long-differenced estimation ($\beta_{\Delta T_{\min}} = +0.075, \beta_{\Delta T_{\max}} = -0.062$). A unidimensional measure folds the opposing effects into a single positive coefficient—growing-season degree-days yield $\beta_{DDs} = +0.036$ on the same panel (online appendix)—because warming in this period is night-biased.

Third, the diurnal reversal on agricultural labour shares scales with the weight of the agricultural sector in the local economy: over two-decade long differences, the night-day divergence is strongest among rural workers ($F = 30.9$) and muted in the urban workforce ($F = 1.9$) (Figure 2). Both rural and urban workforces share a secular decline ($\Delta \bar{y}$) over 1981–2011 ($\Delta \bar{y}^{rural} = -0.046$ rural, $\Delta \bar{y}^{urban} = -0.035$ urban). Within towns (Figure 3), where agriculture is too small to register the divergence, both margins instead *erase* the agricultural-labour decline to a net near-zero (no night-day split, $F = 1.61$) and jointly push down the non-agricultural share ($\beta_{\Delta T_{\max}} = -0.08, \beta_{\Delta T_{\min}} = -0.04, F = 2.78$).

Over long differences, the district-level night-day rural divergence presents a divide between nighttime warming consolidating rural agricultural labour into year-round (main-status) wage work, and daytime warming returning rural labour to the casual seasonal margin—a seasonal return that is sharp in the decadal panel (+3.2 points, Table 2) and directionally present over the long difference (+0.056, $t = 1.58$). Together, the two margins converge at the town-level to reduce non-agricultural employment shares, consistent with a loss in labour productivity and dampened demand for non-agricultural goods and services (Liu, Shamdasani, and Taraz 2023).

Read together, the results present a two-pronged arrest of structural transformation: warmer nights consolidate agricultural labour, warmer days casualize it, and both cut agricultural production.

These empirical facts share a common root: nighttime ($T_{\min}$) and daytime warming ($T_{\max}$) reach the farm through different factors—a warmer night shocks the land, while a warmer day is a neutral productivity loss eroding labour and the underlying land productivity alike. Section 1 synthesizes these findings under a conceptual framework, discussing the assumptions and mechanisms. It maps the daytime warming supply shock

³The Census of India classifies ‘marginal workers’ as those employed under 183 calendar days preceding the year of enumeration—interpreted as ‘seasonal workers’.

to locally rising harvest prices. The framework presents two testable implications—the price asymmetry, that only daytime warming moves the harvest price, and the rural–urban amplitude ordering. An online appendix formalises the full general-equilibrium model and presents robustness decompositions across space, time, and measurement, verifying that the three central empirical facts are robust to specification.

A final piece of suggestive evidence is a near-decadal panel of urban households from three rounds of the National Sample Survey (1987, 1999, 2009) (Ruggles et al. 2024). These household surveys corroborate the mechanism: warmer nights raise urban agricultural labour shares while warmer days raise employment in trade, transport and logistics, mirroring the decadal ‘seasonal’ pattern in urban Censuses. With only three rounds of data, I read this as descriptive evidence.

The distinction between the two temperature shocks is important for welfare. The nighttime shock halts structural transformation by pulling more hands onto the farm; the daytime shock diverts agricultural wage-employment into a low-productivity, informal churn within urban services.

This paper makes three contributions. First, the diurnal decomposition sheds light on the conflicting short- and long-run estimates of heat and labour reallocation in Colmer (2021) and Liu, Shamdasani, and Taraz (2023): it shows what a single degree-day average conceals, that warming nights sort labour within agriculture while warming days reduce formal wage employment, so the direction of change depends on which end of the day warms. Second, a conceptual framework and general equilibrium model (online appendix) show the night/day divergence to be a property of factor bias under gross substitutability of labour and land, carrying an over-identifying restriction—only daytime warming moves the local harvest price ($F = 13.34$), which is consistent with partial market integration. Third, I construct a novel harmonisation of a four-decade panel (1981–2011) linking agricultural statistics (ICRISAT) with employment shares (Census, NSS) at the district and town level, covering a critical economic transition of the largest population in the world.

The rest of the paper proceeds as follows. Section 1 develops the framework. Section 2 describes the data and the empirical approach. Section 3 reports results. Section 4 concludes.

1. Conceptual Framework

This section lays out the economic channels mapping the nighttime and daytime temperature margin to the labour responses estimated in the rest of the paper. The full general-equilibrium model, with proofs, is in the online appendix.

What do nighttime vs daytime temperatures measure? The nighttime minimum and the daytime maximum are the floor and the ceiling of the daily temperature cycle. Holding one fixed, a higher minimum compresses the diurnal range and a higher maximum widens it, so the decomposition speaks to the range—the quantity agronomy ties to grain filling and heat stress—as much as to the level. The two margins act on different inputs. Crops occupy the field around the clock, so a warmer night works on the standing crop and the land in cultivation, raising respiration and the overnight heat load the crop carries (Peng et al. 2004; Welch et al. 2010). Field labour occupies only the daylight, so a warmer day works on the worker, pressing effort against the physiological ceiling of sustained outdoor exertion and lowering the productivity of labour itself (Dunne, Stouffer, and John 2013; Somanathan et al. 2021). A daily average folds the two channels into one number; the analysis keeps them apart. Equivalently, entering the two margins jointly separates the daily *level* (their mean) from the diurnal *range* (their difference): the level moves the farm’s scale—area, output, and self-cultivation—while the range drives the reallocation between wage labour and the seasonal margin and the harvest price, and a degree-day average retains only the level.

Factor-bias from nighttime temperatures. The framework assigns each margin to the factor it acts on: nighttime warming is a *land-biased* shock that raises or lowers the productive services of cultivated land, while daytime warming is a *Hicks-neutral* loss of total factor productivity, consistent with the estimated erosion of agricultural productivity under warming (Ortiz-Bobea et al. 2021). The night margin is distinctive for the factor it lands on rather than the sign of its effect: the nocturnal-respiration and overnight-heat-load channels could in principle either degrade or enhance effective land, and which way is an empirical question the data resolve—humidity-adjusted nights *raise* cereal yields, the signature of net augmentation. The distinction has bite only because labour and land are gross substitutes—the farm can trade one for the other, an elasticity of substitution σ above one. A substantial literature on agricultural production rests on this substitutability: the induced-innovation tradition turns on exactly the land–labour margin (Hayami and Ruttan 1985), classic factor-demand estimates find land and labour to be substitutes (Binswanger 1974), and the production-relations literature for Indian (ICRISAT) village agriculture builds on the same substitutability (Binswanger and Rosenzweig 1986), while cross-country CES estimates cluster around unity and vary with method and setting (Mundlak, Butzer, and Larson 2012). I do not estimate σ ; I maintain $\sigma > 1$ as the empirically relevant case (online appendix). What disciplines the assignment is the joint pattern the data return—agricultural labour rising under warmer nights while area and output contract and the price stays flat, a combination no merely beneficial neutral shock delivers. When a warmer night *augments* the quality of an acre, a farm that substitutes readily reworks its land-to-labour mix and works the land harder, hiring *more* field labour

rather than less; a degrading night shock would instead lower labour demand. The sign of the shock to the factor sets the direction of the labour response, independent of yields—consistent with findings from Bustos, Caprettini, and Ponticelli (2016) in Brazil, where land-augmenting maize pulled labour into agriculture and labour-saving soy released it. In this paper, the diurnal heat cycle supplies the two biases. A neutral productivity loss lands on no factor in particular: it leaves the land-to-labour ratio intact and shrinks the scale of the operation, pulling down the marginal product of labour and the field wage and, where the food market is tight, raising the harvest price. The two margins therefore move farm-labour demand in *opposite* directions—warmer nights raise it, warmer days lower it. Under Cobb–Douglas ($\sigma = 1$) a land-augmenting shock folds into total factor productivity, the two channels collapse, and a daily average loses nothing.

The seasonal employment ‘sink’. Workers divide across four states—agricultural wage labour, self-cultivation, non-agricultural work, and casual seasonal work—whose shares sum to one, so the margins move as one offsetting system, and the casual seasonal pool is the flexible residual that clears the reallocation. When nighttime warming raises the demand for farm labour, the farm hires: the wage-labour share rises and draws from the cultivators who give up their marginal plots and the casual workers who attach to the harvest—the mark of a commercialising farm that retains and reorganises its labour. When daytime warming collapses the marginal product, the wage-labour market gives way and returns workers to the seasonal margin. Self-cultivation falls under both margins, as marginal plots go unworked.

The price test for market integration. I test the decomposition’s openness rather than assume it. Only a productivity-driven (daytime) output loss raises the local harvest price; the nighttime contraction leaves it flat. That asymmetry marks a district whose food price is set largely by trade, not by local scarcity (partial market integration). The online appendix embeds this in a general-equilibrium closure spanning open and closed food markets and shows that the over-identifying price restriction places India’s districts on the integrated side; a fully closed subsistence economy would instead invert the labour orientation. Partial integration lets a daytime output loss raise the local harvest price, which is why only the daytime margin moves it.

Intepreting the rural-urban channels. Rural and urban populations share the nighttime sign and differ in amplitude; the urban daytime response is a statistical zero. The factor-bias divergence is a mechanism that operates inside the agricultural sector; its footprint in local employment shares therefore scales directly with that sector’s weight in the local economy. Where agriculture is the dominant sector—rural district populations—the

reallocation registers sharply, amplified further by the depth of the casual-labour margin that absorbs it. Where agriculture is a minor sector—towns—the within-agriculture divergence attenuates, and the dominant margin becomes the sector’s own heat response: a shared loss of capacity for outdoor informal work that contracts non-agricultural employment under both margins. This is consistent with the framework’s supply channel and with dampened local demand for non-farm goods and services under heat (Liu, Shamdasani, and Taraz 2023; Nath 2025). The framework belongs to the structural-transformation tradition (Matsuyama 1992; Kongsamut, Rebelo, and Xie 2001; Gollin and Rogerson 2014; Bustos, Caprettini, and Ponticelli 2016; Barrett, Ortiz-Bobea, and Pham 2023); the online appendix develops the full general-equilibrium model—primitives, the gross-substitutes assumption, the price closure, and the lemma and propositions behind the predicted signs, with all proofs. These signs guide the analysis rather than serving as a structure to be estimated; the next sections test them.

2. Data and Empirical Approach

2.1. Data Sources

I harmonise four datasets to link climate, agricultural production and structural transformation: (1) daily gridded weather data from the IMD, (2) district agricultural data (ICRISAT), (3) district and town level tables from the Indian Census, and (4) urban household-level employment data from the NSS. These datasets span four decades (Census, IMD, ICRISAT), with a shorter window from the NSS (1987, 1999, 2009).

Indian Meteorological Department. Gridded data were obtained from the Indian Meteorological Department (IMD) on air temperature (1° resolution) and precipitation (0.25° resolution) over 1951–2023. Consistent with the approach in Liu, Shamdasani, and Taraz (2023), daily observations are averaged over growing seasons of the past 10 years. The growing season marks the months of June–February, which captures cropping cycles of rice in the monsoon (kharif) and wheat in the winter (rabi) seasons. The baseline exposure measures are the growing-season means of the daily minimum and maximum surface-air temperatures, T_{min} and T_{max} , which separate nighttime from daytime warming. For comparison I also construct degree days (DDs), a scalar that rescales and sums agriculturally relevant temperatures between 8 and 32 °C (Blakeslee and Fishman 2015; Schlenker, Hanemann, and C. 2006); degree days average over the diurnal cycle and so collapse the two margins into one.

$$(1) \quad DDs = \sum_d D(T_d) ; \text{ where } T_d = \frac{T_{min,d} + T_{max,d}}{2}$$

$$(2) \quad D(T) = \begin{cases} 0, & \text{if } T \leq 8^\circ\text{C} \\ T - 8, & \text{if } 8^\circ\text{C} < T \leq 32^\circ\text{C} \\ 24, & \text{if } T > 32^\circ\text{C} \end{cases}$$

As a humidity-adjusted robustness measure I also construct growing-season wet-bulb minimum and maximum temperatures following Stull (2011); these are derived from T_{min} and T_{max} alone—no external humidity series is used (the construction is detailed in the supplemental data appendix)—and enter the wet-bulb specifications reported there.

Census of India. The core dataset is sourced from the District Census Handbooks (DCHBs) of India. Specifically, I digitise the Primary Census Abstract (PCA) tables, covering the universe of Indian towns from 1981 to 2011.⁴ A Census town meets three criteria: (1) a minimum population of 4,000, (2) at least 75% of the male working population in non-agricultural activity, and (3) a population density above 400 persons per sq. km. I construct a panel of 2,969 towns that appear consistently across all four decades, so that the analysis reflects intensive-margin adjustments rather than the reclassification of urban areas. For comparability over time, occupations are aggregated into four mutually exclusive categories: cultivators (landowners), agricultural labourers (wage workers), non-agricultural workers, and marginal workers. The “marginal” category cannot be assigned cleanly to agriculture or non-agriculture, but it lets me track the contractual *quality* of work. The 2001 Census enumerated marginal work more completely than 1991; the headline results are identified off the consistently defined main-worker categories, and a per-capita denomination immune to the change reproduces them (online appendix). The harmonised district panel covers roughly 300 districts across twenty major states on 2011-consistent boundaries; eight north-eastern states lack a usable 1981 identifier and are excluded (online appendix).⁵

ICRISAT. The ICRISAT District Level Database (DLD) tracks agricultural inputs and production in India from 1966 to the present. I harmonise districts to 2011-Census consistent boundaries, constructing an annual panel of cultivated area, production, yields and harvest prices for key grains: rice, wheat, maize, sorghum, barley and millets. I construct the mean of these measures over 10 years preceding each Census round, e.g. over the years 1971–1980 for the 1981 Census. Thus, the results from the temperature

⁴The town digitisation builds on data assembled for Blakeslee et al. (2023).

⁵Of the 300 districts on 2011-consistent boundaries, 299 admit a computable long difference in the agricultural wage-labour share. The regressions absorb state fixed effects, dropping six single-district states and union territories as singletons (Chandigarh, Delhi, Dadra & Nagar Haveli, Lakshadweep, Puducherry, Andaman & Nicobar) for $N = 293$. The residence-specific panels lose two more each—two fully urban districts (Mumbai Suburban, Chennai) have no rural population and two fully rural ones (Lahul & Spiti, Kinnaur) no urban—so that share is undefined ($N = 291$); the log-worker columns retain 293.

regressions compare contemporaneous horizons of 10-year windows with climate data. The DLD covers staple grains only; horticulture and cash crops—which have grown rapidly and differ in labour intensity, irrigation, and market structure—lie outside it, so the agricultural-economy results speak to staple-grain production and the broader composition of Indian agriculture is left to future work. The Census labour reallocation, however, survives this composition shift: it is strongest in districts that stayed in staple grains and muted where agriculture diversified into high-value crops (online appendix).

National Sample Surveys. I use three rounds of the National Sample Survey (NSS) Employment and Unemployment Schedules (1987, 1999, and 2009), harmonised by IPUMS (Ruggles et al. 2024). While the Census measures the volume of labour reallocation, it lacks a granular view of the destination sectors within non-agriculture under its 1981 definitions. Restricting the NSS sample to urban households serves as an independent benchmark to the Census urban and town-level data. The NSS urban sampling frame is separate from Census towns (it retains more peri-urban agriculture), and provides complementary evidence of the divergence in agricultural wage-labour shares. The data additionally shed light on the destination of heat-displaced labour—casual, informal work rather than stable formal employment in manufacturing or services.

2.2. Empirical Approach

The identification takes two approaches: a fixed effects panel estimation (3), and a long differences approach (4), consistent with Dell, Jones, and Olken (2014), Burke and Emerick (2016) and Liu, Shamdasani, and Taraz (2023). Let Y_{irt} be the outcome of interest (e.g., share of workers in agriculture) for district or town i , within state r , observed at time t . $\mathbf{T}_{irt}$ denotes a vector of temperature measures, either degree days $\mathbf{T} = \{DDs\}$ or the diurnal surface-air pair $\mathbf{T} = \{T_{min}, T_{max}\}$. All specifications control for total precipitation $\mathbf{P}_{it}$ and fit a state-specific linear time trend.

$$(3) \quad Y_{irt} = \beta \mathbf{T}_{irt} + \phi \mathbf{P}_{irt} + \mu_i + \lambda_t + \gamma_r \cdot t + \varepsilon_{irt}$$

$$(4) \quad \Delta Y_{ir} = \beta \Delta \mathbf{T}_{ir} + \phi \Delta \mathbf{P}_{ir} + \gamma_r + \varepsilon_{ir}$$

For temporal parity, I construct the gridded temperature and rainfall measures as a moving average over the growing seasons of the preceding ten years. Decadal climate is the inverse-distance-squared-weighted average of IMD grid points surrounding the district (within 200 km) or town (within 100 km) centroid—radii chosen to match the

distance at which the residual spatial correlation of the outcomes decays (roughly 150–200 km for districts and 100 km for towns). Because the weights fall off as the squared distance, grid points beyond roughly 100 km contribute negligibly, so the assignment is insensitive to the exact cutoff.

Equation 3 is a district- or town-decade panel with two-way fixed effects and a state-specific linear trend; because the climate regressor is itself a ten-year moving average, I read its coefficients as the *decadal* response of structural transformation to heat. Equation 4 averages the outcome Y and covariates X over two endpoints—1981–91 and 2001–11—and regresses $\Delta Y = Y_2 - Y_1$ on $\Delta X = X_2 - X_1$. This long-differenced estimator recovers the effect of a sustained increase in T on the outcome trend ΔY , measured against its secular trend $\Delta \bar{Y}$ over the sample. Both estimators use the same decadal smoothed climate; the long difference adds one further decade of horizon and absorbs unobserved state-level trajectories ($\gamma_r \cdot t$). I treat it as a robustness design—confirming that the decadal-panel result survives a different fixed-effects and averaging structure that strips decade-common shocks—rather than as a distinct long-run or adaptation-inclusive estimand.⁶

I enter T_{min} and T_{max} simultaneously and read each coefficient as the partial effect of one margin holding the other—and rainfall—constant. Inference faces two distinct dependencies. The first is the decadal *persistence* of the climate regressor (a ten-year moving average), which inflates rejection rates unless the estimator accounts for within-unit serial correlation. This choice is disciplined by Monte Carlo simulations (online appendix), where I regress the key outcomes on a placebo climate series—the real series randomly permuted across districts—preserving decadal persistence. I select the preferred panel estimator for equation 3 as the specification that rejects the null in roughly 5% of simulations. I find that unadjusted and heteroskedasticity-robust standard errors over-reject sharply, whereas clustering by district with state-specific linear trends restores the nominal 5% size.

The second dependency is spatial—a persistent, strongly associated feature of Indian diurnal-temperature data (Bhattacharjee and Bose 2024); because the permutation reassigns whole series across units it breaks the regressor’s spatial structure, so the Monte Carlo does not speak to spatial correlation, which I address separately with Conley spatial-HAC standard errors (Conley 1999) reported alongside the clustered errors on the panel exhibits. The online appendix sweeps the Conley radius to 200 km: the night/day

⁶Lemoine (2021) shows that long-difference and panel estimators of climate impacts are identified by overlapping variation in the realised temperature series (here decadal averaged), so a long difference need not isolate differential rates of climate change or a separate long-run adaptation response, and agreement between the two designs reflects robustness to the estimating structure rather than evidence on adaptation dynamics. Identification here therefore rests on the within-unit, simultaneous T_{min} -versus- T_{max} contrast in the decadal panel; the ten-year smoothing and the long difference serve as robustness rather than as a short-run/long-run decomposition.

equality rejection is stable at every radius ($F = 12.4$ at 200 km), with the daytime labour coefficient thinning to the ten-percent level at the widest radius.

The simultaneous estimator addresses two identification challenges. First, because T_{min} and T_{max} are correlated ($\rho \approx 0.78$ across districts), entering either alone would induce omitted-variable bias; entering them together isolates their partial effects and identifies a narrowing versus widening of the local diurnal range. The district and decade fixed effects and state trends absorb most of this co-movement, so the *estimating* variation is close to orthogonal (within-transformed $VIF \approx 1$; the online appendix) and the partial signs are well-identified. By Frisch–Waugh–Lovell, the coefficients are determined exactly from this residual variation, so the within correlation governs whether the two margins can be separated—and within, they are all but orthogonal. Absorbing state-specific trends accounts for time-varying state policies and compares districts and towns within their local economic geography under similar baseline climate trajectories. Correlograms of the residuals confirm that the spatial correlation of these heat shocks in the key outcomes decays within roughly 100 km for towns and 150–200 km for districts.

3. Results

Tables 1 and 2 decompose the effects of warming nights and days on the district agricultural economy: Table 1 reads down the supply chain for the major cereals—rice, wheat, maize, sorghum, barley, and millets—and Table 2 reports the within-district reallocation of labour. The nighttime minimum and the daytime maximum carry opposite signs across the price and labour outcomes, consistent with the theoretical framework.

Both margins contract the farm. Cultivated land falls by a statistically identical amount across the two margins (a 1°C rise cuts log area by 0.22 either way, $F = 0.01$), while production falls by comparable amounts (-0.15 at night, -0.22 by day, $F = 0.8$). I read the milder nighttime loss as labour intensification; the significant nighttime rise in wage labour adds output back through the production identity, cushioning the extensive-area loss, while daytime labour-shedding compounds it. The two margins diverge on the harvest price. The daytime maximum raises the log harvest price by 0.13 while the nighttime minimum holds it flat, a split sharp at $F = 13.34$; the field wage edges down under both (-0.12 by day, -0.04 at night, too close to separate). Daytime warming thus acts as a supply shock, the loss of land and grain passing into a dearer harvest price; the framework reads the nighttime contraction as concentrated on marginal land whose output is more nearly consumed on-farm, passing little into the marketed surplus that sets that price (online appendix). Cereal yields move little under either margin in the dry-bulb baseline; the augmentation the labour response implies surfaces only under humidity-adjusted temperatures, where night yields rise outright (online appendix). The aggregate also blends opposing grain-level responses, with heat-tolerant coarse cereals

TABLE 1. Panel estimates of diurnal warming on the district agricultural economy

	Log Levels (All Grains)				
	Cultivated Area (1)	Production (2)	Harvest Price Index (3)	Field Wage (4)	Yield Index (5)
T_{min}	-0.218 (-4.76) [-4.76]	-0.145 (-2.26) [-2.25]	-0.042 (-1.36) [-1.36]	-0.038 (-0.62) [-0.63]	0.051 (1.27) [1.28]
T_{max}	-0.225 (-4.01) [-4.02]	-0.221 (-3.47) [-3.46]	0.130 (3.81) [3.77]	-0.119 (-1.70) [-1.76]	-0.046 (-0.97) [-0.97]
Observations	1,243	1,243	1,191	1,115	1,211
Dep Var. Mean	5.45	5.77	5.66	2.86	7.14
F-stat ($T_{min} = T_{max}$)	0.01	0.80	13.34	0.74	2.63
p -value	0.905	0.373	<0.001	0.391	0.106
Rainfall Control	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y

Notes: Log cultivated area, log production, a log harvest-price index, the log field wage, and an area-weighted yield index for the major Indian cereals (rice, wheat, barley, maize, millets, sorghum), from the ICRISAT district panel. T_{min} and T_{max} are the growing-season (June–February) surface-air minimum and maximum, averaged over the preceding decade; all specifications control for total precipitation over the same window. The data form a decadal panel (1981–2011) on 2011 Census-consistent district boundaries, with district and decade two-way fixed effects and a state $\times$ decade linear trend. Parentheses report t -statistics from district-clustered standard errors; brackets report t -statistics from Conley spatial-temporal HAC standard errors (50 km, four decadal lags); bold coefficients indicate $|t| > 2$. The F -statistic and its p -value test $T_{min} = T_{max}$.

TABLE 2. Panel estimates of diurnal warming on district labour reallocation

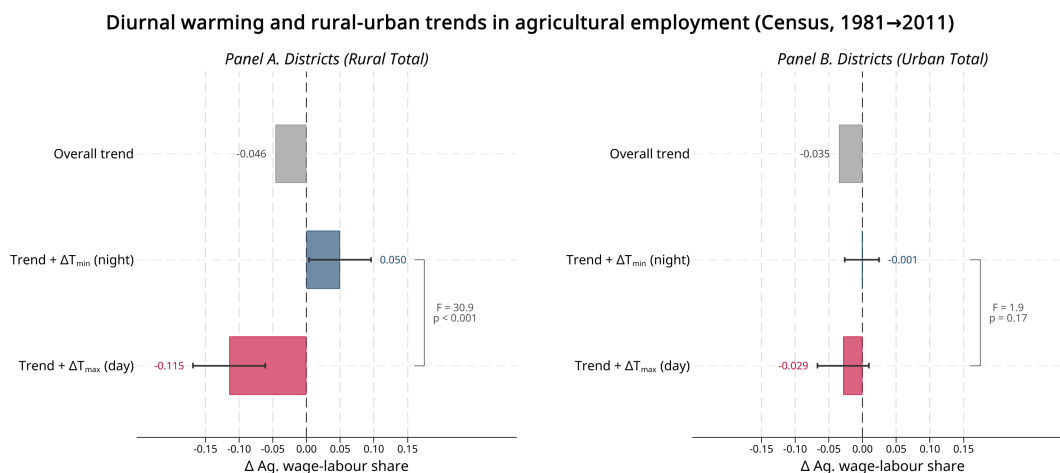
	Share of Workers				Log Workers (5)
	Ag. Labour (1)	Cultivator (2)	Seasonal (3)	Non- Agriculture (4)	
T_{min}	0.044 (4.59) [4.65]	-0.028 (-2.21) [-2.23]	-0.033 (-2.83) [-2.83]	0.017 (1.46) [1.48]	0.175 (1.99) [2.02]
T_{max}	-0.026 (-2.76) [-2.77]	-0.010 (-0.83) [-0.84]	0.032 (2.45) [2.43]	0.004 (0.32) [0.32]	-0.017 (-0.20) [-0.21]
Observations	1,187	1,187	1,187	1,187	1,187
Dep Var. Mean	0.18	0.32	0.17	0.32	13.45
F-stat ($T_{min} = T_{max}$)	24.91	0.85	14.66	0.54	2.36
p -value	<0.001	0.356	<0.001	0.462	0.125
Rainfall Control	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y

Notes: District shares of total workers by category—agricultural wage labour, cultivators, the casual seasonal margin, and non-agriculture—and, in the last column, the log number of district workers, from the Census district panel. Outcomes, sample, fixed effects, and inference follow Table 1: a decadal panel (1981–2011) with district and decade fixed effects and a state $\times$ decade linear trend, a precipitation control, district-clustered t -statistics in parentheses over Conley spatial-temporal HAC t -statistics in brackets, and bold coefficients for $|t| > 2$. The F -statistic and its p -value test $T_{min} = T_{max}$.

gaining under warmer nights while wheat loses, so the cereal index is a crop-composition average (online appendix).

Table 2 reports the labour reallocation. A warmer night raises the agricultural wage-labour share by 4.4 points and draws that labour off self-cultivation (−2.8) and the casual seasonal margin (−3.3); a warmer day reverses the flow, cutting the wage-labour share by 2.6 points and returning 3.2 points to the seasonal margin. The two margins separate sharply ($F = 24.9$ on wage labour, $F = 14.7$ on the seasonal share), while non-agricultural employment holds flat under both ($F = 0.5$), so the district reshuffles labour within agriculture. The extensive count rises at night, a warmer minimum lifting the log number of district workers by 0.18, at the margin of significance. The appendix framework grounds the split: nighttime warming acts on land as a land-biased shock, and because labour and land are gross substitutes it raises the demand for field labour directly, while daytime warming is a neutral productivity loss that shrinks the operation and sheds labour to the seasonal pool.

FIGURE 2. Long-differenced estimates of diurnal warming on agricultural labour trends



Notes: Long-difference estimates (1981–2011) of a 1°C rise in growing-season nighttime (T_{min}) and daytime (T_{max}) surface-air temperature on the district agricultural wage-labour share, estimated separately for the rural and urban populations of each district. Grey bars give the secular trend (the mean change in the share); each coloured bar adds the warming coefficient to that trend ($\bar{y} + \hat{\beta}$), with 95% confidence intervals (spikes); the brace reports the night-versus-day equality test ($\Delta T_{min} = \Delta T_{max}$). A coloured bar whose interval clears its grey trend bar is a statistically significant warming effect. All specifications control for the long-run change in precipitation and include state fixed effects; standard errors are clustered at the district level. The full total/rural/urban coefficients behind this figure are in Appendix Table A3.

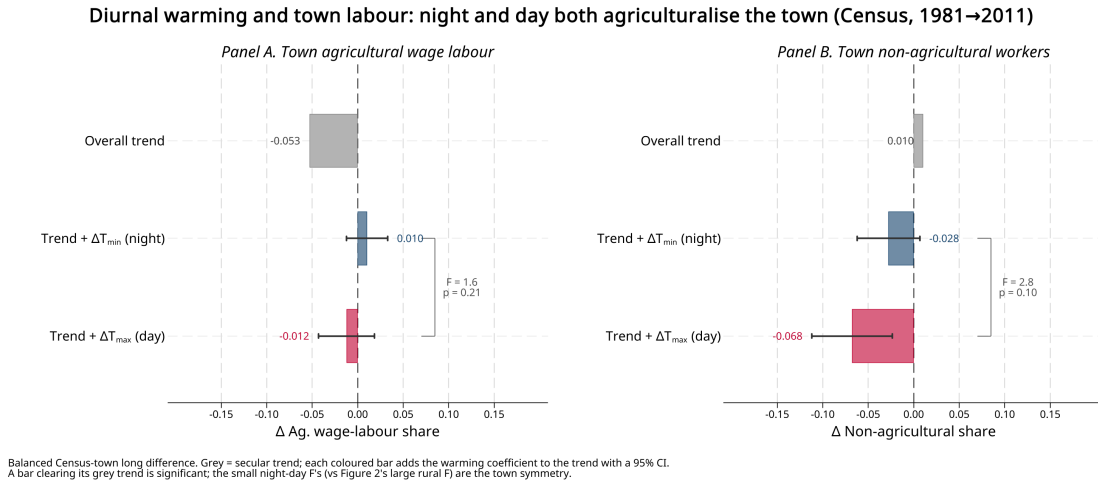
Figure 2 carries the labour decomposition to its two-decade horizon and splits each district into its rural and urban populations. The long difference averages each district's two Census endpoints—1981 and 1991 against 2001 and 2011—and differences them, stripping decade-specific noise to isolate the change in climate against the change in labour. The grey bars set the stage: the agricultural wage-labour share fell across rural

and urban India over these two decades, the secular structural transformation drawing labour off the land.

Warming bends that trend, and the two margins bend it in opposite directions. A warmer night raises the rural agricultural wage-labour share by 0.096 and turns its -0.046 secular decline into a $+0.050$ rise: nighttime warming hoards labour on the farm against the structural trend. A warmer day lowers the same share by 0.068, deepening the decline to -0.114 : daytime heat pushes a rural agricultural exit. The opposition is sharp and holds over the two-decade horizon.

The urban populations of the same districts barely move. A warmer night lifts the urban agricultural share by 0.034—a third of the rural response—and a warmer day leaves it flat ($+0.006$, insignificant). The reversal is a rural phenomenon; the urban populations of these districts absorb little of it. The framework reads this as amplitude (online appendix): the diurnal response scales with the agricultural share of local employment, deep in rural districts and thin in urban ones, so the same factor-bias force registers sharply in rural areas and faintly in the town. The online appendix reports the full rural and urban decomposition behind the figure—all four worker shares and log total workers. District populations move in the same direction as workers, with the urban population rising under nighttime warming at the ten-percent level (online appendix).

FIGURE 3. Diurnal warming and town labour reallocation over the long difference



Notes: Long-difference (1981–2011) estimates of a 1°C rise in growing-season nighttime (ΔT_{min}) and daytime (ΔT_{max}) surface-air temperature on Census-town worker shares, for the balanced panel of towns. Panel A is the agricultural wage-labour share; Panel B the non-agricultural share. Grey bars give the secular trend (the mean change in the share); each coloured bar adds the warming coefficient to that trend ($\bar{y} + \hat{\beta}$), with 95% confidence intervals; the brace reports the night-versus-day equality test ($\Delta T_{min} = \Delta T_{max}$). A coloured bar whose interval clears its grey trend bar is a statistically significant warming effect, read against Figure 2's rural asymmetry. All specifications control for the long-run change in precipitation and include state fixed effects; standard errors are clustered at the district level. The underlying coefficients—all four worker shares and log town workers—are in Appendix Table A4.

Figure 3 carries the decomposition to the Census town and closes the loop on the central finding. Where agriculture is the dominant sector—rural district populations—the two margins split sharply (Figure 2). Where it is a minor one—the urban populations of those same districts, and the towns here—the wage-labour split collapses, and warming does something else. It does two things at once. First, both margins *erase* the town agricultural-labour decline: the -0.053 secular fall is offset to a net near-zero ($+0.010$ under warmer nights, -0.012 under warmer days), with no day/night divergence on the wage-labour share ($F = 1.6$)—though the seasonal margin retains a residual night–day split ($F = 4.1$, $p = 0.04$; Appendix Table A4), a faint echo of the district pattern. This is the same net erasure the urban district shows in Figure 2, where nighttime warming lifts the -0.035 urban decline to -0.001 : the positive nighttime coefficients do not push the town toward agriculture so much as cancel its retreat from it. Second—and this is what the town, unlike the urban district, makes visible—the non-agricultural share, which drifted *up* by about a point over these decades, is pushed down by both margins: warmer days cut it sharply (-0.078) and warmer nights more mildly (-0.038), too alike to separate ($F = 2.8$). The two margins move it together, so the daily *level* and the diurnal *range* of warming both contract town non-agricultural employment. This town de-urbanisation—the erased farm-labour decline alongside a squeezed non-farm sector—mirrors Blakeslee et al. (2023), where irrigation likewise raises rural agriculture and contracts the town non-farm sector. The pattern is directionally the same in the full (unbalanced) town panel, where the non-agricultural drag stays symmetric across the two margins though muted, alongside a residual nighttime pull onto town agriculture; the balanced long difference, robustness across the town sample, and the suggestive (imprecise) in-migration are in the online appendix.

A final, suggestive question is where the displaced labour lands. Matched to three rounds of the National Sample Survey, the urban data reproduce the district mechanism on its matched margin—a warmer night raises the urban agricultural wage-labour share ($+0.060$, $p < 0.01$)—and hint that daytime warming diverts labour into informal trade and transport rather than formal employment (online appendix). With only three survey rounds I read this as suggestive and leave it to future work.

The online appendix confirms the decomposition under alternative measures: a single degree-days measure collapses the opposition into one muted coefficient; and wet-bulb temperatures sharpen the labour split while humid nights raise cereal yields outright ($+0.15$), the land-augmenting signature ($B' > 0$) absent from the dry-bulb baseline, where the night margin is on net contractionary.

4. Conclusion

This paper asks which way warming temperatures move agricultural labour, and finds a diurnal divergence in the results. To my knowledge, this paper provides the first reduced-form evidence and a conceptual framework for understanding the economic channels of diurnal warming — shifts in nighttime and daytime temperatures — on structural transformation. The framework carries a single over-identifying restriction: only the daytime margin moves local harvest prices. The data confirm this asymmetry ($F = 13.34$ testing equality $\beta_{T_{\min}} = \beta_{T_{\max}}$ on prices), providing an empirical grounding for partial market integration, independent from labour reallocation. The framework is a guide on the signs rather than magnitudes of the empirical estimates from a panel and long-differenced approach, which test these comparative statics. Though the diurnal divergence is robust to a battery of stress tests, the empirical approach has a few important nuances and limitations.

First, while multicollinearity of the within-variation (ρ) is a concern for identification under wet-bulb temperatures $\rho \approx 0.84$, in the surface-air baseline the two margins are within-correlated at only $\rho \approx -0.09$ (online appendix). Twoway fixed effects and state trends absorb almost all of that co-movement, leaving the identifying variation near-orthogonal (within-transformed VIF ≈ 1 , condition number ≈ 1.1). Both $T_{\min}$, $T_{\max}$ carry their sign on agricultural labour shares (+/-, respectively) when specified independently with and without rainfall controls in both panel and long differenced estimations. The within-correlation widens standard errors without drastically altering the magnitude and sign of the point estimates $\beta_{T_{\min}}, \beta_{T_{\max}}$.

Second, the wet-bulb temperature specification is a deterministic nonlinear approximation using the raw dry-bulb $T_{\min}$ and $T_{\max}$ (online appendix), mechanically compressing the diurnal temperature range (DTR). This makes the wet-bulb pair highly collinear relative to the dry-bulb baseline (within- $\rho \approx 0.84$, within-transformed VIF ≈ 3.5 ; the online appendix).

Third, specified jointly, $T_{\min}$ and $T_{\max}$ span the same two-dimensional regressor space as the daily mean and the diurnal range (DTR), so the diurnal decomposition mirrors temperature level vs range effects, uninfluenced by the chosen basis. A mean-range reparametrisation provides supporting evidence that the DTR carries an independent channel on the labour and price margins (online appendix).

Fourth, diurnal warming is a *local* environmental change, compressed or widened by irrigation, aerosols, and land use. These local changes are arguably endogenous to the production function as emphasised by Barrett, Ortiz-Bobea, and Pham (2023). Thus, the diurnal margins are not randomly assigned. Irrigation expansion and other drivers of range compression raise nighttime and lower daytime temperatures in the same agrarian belts where labour reallocates. Three features in the data sample address

this confounding: cultivated area contracts identically under both margins, where an irrigation-driven night signal would instead expand it; the nighttime yield dividend persists within the historically irrigated belt (online appendix); and measurement drift in gridded nighttime temperatures near growing settlements would bias the nighttime coefficient toward zero, rejected by the empirical estimates. Moreover, the asymmetry is rooted in atmospheric science: the shallow nocturnal boundary layer amplifies the nighttime response to any change in energy-balance, upstream of local land use (Davy et al. 2016).

Finally, the conceptual framework maintains labour and land as gross substitutes ($\sigma > 1$). Though it is grounded in the agricultural-production literature (Hayami and Ruttan 1985; Binswanger 1974), I do not directly estimate this assumption. The factor-bias is sufficiently derived from non-unit elasticity: under a Cobb–Douglas production function, the land shock folds into neutral productivity (online appendix). The consistent signs on labour, area, output and prices discipline this assignment. Nonlinearity and spatial sorting, held fixed by the linear framework, are left for future research.

A natural extension of the framework’s prediction is through India’s own colonial-era and early post-Independence agricultural records, which provide a test for whether a fully closed subsistence economy without buffers inverts the night-day pattern. The pattern is consistent in itself: warmer nights hold labour on the farm without the price and wage distress that warmer days bring, while daytime heat restrains the non-agricultural transition far more strongly. Given the two margins carry opposite signs on agricultural labour, the trajectory of the diurnal temperature range matters as much as the level. Where nights are warming faster than days—compressing the range (Karl et al. 1991)—the decomposition implies agricultural consolidation and commercialisation. Where days outpace nights and the range widens (Zhong et al. 2023), warming implies the double squeeze of farm-labour casualisation and stalled urban structural transformation.

A back-of-the-envelope calculation—comparing each district’s trailing-decade climate at the 2011 and 1981 Censuses—suggests growing-season nights and days warmed by 0.45° and 0.37°C on average over 1981–2011, but the balance is regional—in the districts where the range compressed, nights ran ahead of days ($+0.71$ versus $+0.17^\circ\text{C}$), and where it widened, days led ($+0.54$ versus $+0.21^\circ\text{C}$). Applying panel estimates from Table 2, warming in the range-compressing districts implies a 2.7-point rise in the agricultural wage-labour share against an 18% base. The range-widening $T_{\max}$ warming implies a slight decline and a 1.0-point shift onto the casual seasonal margin. Night and day warming, in equivalent magnitudes, sort local agricultural labour in opposite directions. Under extreme daytime heat, the exit from agricultural labour into informal seasonal employment reflects a physiologically constrained churn for survival, rather than a sustained pathway to structural change.

References

- Barrett, Christopher B., Ariel Ortiz-Bobea, and Trinh Pham. 2023. "Structural Transformation, Agriculture, Climate, and the Environment." *Review of Environmental Economics and Policy* 17 (2): 195–216. 10.1086/725319.
- Bhattacharjee, Madhuchhanda and Arup Bose. 2024. "Spatio-temporal Patterns of Diurnal Temperature: A Random Matrix Approach I—Case of India." <https://arxiv.org/abs/2404.11747>. ArXiv preprint.
- Binswanger, Hans P. 1974. "The Measurement of Technical Change Biases with Many Factors of Production." *American Economic Review* 64 (6): 964–976.
- Binswanger, Hans P. and Mark R. Rosenzweig. 1986. "Behavioural and Material Determinants of Production Relations in Agriculture." *Journal of Development Studies* 22 (3): 503–539. 10.1080/00220388608421994.
- Blakeslee, David, Aaditya Dar, Ram Fishman, Samreen Malik, Heitor S. Pellegrina, and Karan Singh Bagavathinathan. 2023. "Irrigation and the Spatial Pattern of Local Economic Development in India." *Journal of Development Economics* 161: 102997. 10.1016/j.jdeveco.2022.102997.
- Blakeslee, David S. and Ram Fishman. 2015. "Weather Shocks, Agriculture, and Crime: Evidence from India." 10.2139/ssrn.2428249. [Online; accessed 23. Nov. 2024].
- Burke, Marshall and Kyle Emerick. 2016. "Adaptation to Climate Change: Evidence from US Agriculture." *American Economic Journal: Economic Policy* 8 (3): 106–40. 10.1257/po1.20130025.
- Bustos, Paula, Bruno Caprettini, and Jacopo Ponticelli. 2016. "Agricultural Productivity and Structural Transformation: Evidence from Brazil." *American Economic Review* 106 (6): 1320–65. 10.1257/aer.20131061.
- Chen, Shuai, Xiaoguang Chen, and Jintao Xu. 2016. "Assessing the impacts of temperature variations on rice yield in China." *Climatic Change* 138 (1–2): 191–205. 10.1007/s10584-016-1707-0.
- Colmer, Jonathan. 2021. "Temperature, Labor Reallocation, and Industrial Production: Evidence from India." Working paper, SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3900866.
- Conley, Timothy G. 1999. "GMM estimation with cross sectional dependence." *Journal of Econometrics* 92 (1): 1–45. 10.1016/s0304-4076(98)00084-0.
- Das, Moumita and Anirban Sanyal. 2026. "Adaptation in Motion: Temporary Migration under Heat Stress." Job market paper, University of California, Santa Cruz.
- Dasgupta, Aparajita and Devvrat Raghav. 2024. "Rural Roads, Climate Change, and the Dynamics of Structural Transformation: Evidence from India." Economics Discussion Paper 122, Ashoka University.
- Davy, Richard, Igor Esau, Alexander Chernokulsky, Stephen Outten, and Sergej Zilitinkevich. 2016. "Diurnal asymmetry to the observed global warming." *International Journal of Climatology*. 10.1002/joc.4688.
- Dell, Melissa, Benjamin F. Jones, and Benjamin A. Olken. 2014. "What Do We Learn from the Weather? The New Climate-Economy Literature." *Journal of Economic Literature* 52 (3): 740–98. 10.1257/jel.52.3.740.
- Dunne, J. P., R. J. Stouffer, and J. G. John. 2013. "Reductions in labour capacity from heat stress

- under climate warming." *Nature Climate Change* 3 (6): 563–566. 10.1038/nclimate1827.
- Easterling, David R., Briony Horton, Philip D. Jones, Thomas C. Peterson, Thomas R. Karl, David E. Parker, M. James Salinger, Vyacheslav Razuvayev, Neil Plummer, Paul Jamason, et al. 1997. "Maximum and Minimum Temperature Trends for the Globe." *Science* 277 (5324): 364–367. 10.1126/science.277.5324.364.
- Gollin, Douglas and Richard Rogerson. 2014. "Productivity, transport costs and subsistence agriculture." *Journal of Development Economics* 107: 38–48. 10.1016/j.jdeveco.2013.10.007.
- Hayami, Yujiro and Vernon W. Ruttan. 1985. *Agricultural Development: An International Perspective*. Revised ed. Baltimore, MD: Johns Hopkins University Press.
- Herrendorf, Berthold, Richard Rogerson, and Akos Valentinyi. 2014. "Growth and Structural Transformation." In *Handbook of Economic Growth*, vol. 2, pp. 855–941. Elsevier. 10.1016/B978-0-444-53540-5.00006-9.
- IPCC. 2021. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press. 10.1017/9781009157896.
- Jessoe, Katrina, Dale T. Manning, and J. Edward Taylor. 2017. "Climate Change and Labour Allocation in Rural Mexico: Evidence from Annual Fluctuations in Weather." *The Economic Journal* 128 (608): 230–261. 10.1111/eoj.12448.
- Karl, Thomas R., Philip D. Jones, Richard W. Knight, George Kukla, Neil Plummer, Vyacheslav Razuvayev, Kevin P. Gallo, Janette Lindseay, Robert J. Charlson, and Thomas C. Peterson. 1993. "A New Perspective on Recent Global Warming: Asymmetric Trends of Daily Maximum and Minimum Temperature." *Bulletin of the American Meteorological Society* 74 (6): 1007–1023. 10.1175/1520-0477(1993)074<1007:ANPORG>2.0.CO;2.
- Karl, Thomas R., George Kukla, Vyacheslav N. Razuvayev, Michael J. Changery, Robert G. Quayle, Richard R. Heim, David R. Easterling, and Cong Bin Fu. 1991. "Global warming: Evidence for asymmetric diurnal temperature change." *Geophysical Research Letters* 18 (12): 2253–2256. 10.1029/91GL02900.
- Kongsamut, Piyabha, Sergio Rebelo, and Danyang Xie. 2001. "Beyond Balanced Growth." *Review of Economic Studies* 68 (4): 869–882. 10.1111/1467-937X.00193.
- Lemoine, Derek. 2021. "Estimating the Consequences of Climate Change from Variation in Weather." NBER Working Paper 25008, National Bureau of Economic Research, Cambridge, MA. Revised May 2021.
- Lewis, W. Arthur. 1954. "Economic Development with Unlimited Supplies of Labour." *The Manchester School* 22 (2): 139–191. 10.1111/j.1467-9957.1954.tb00021.x.
- Liu, Maggie, Yogita Shamdasani, and Vis Taraz. 2023. "Climate Change and Labor Reallocation: Evidence from Six Decades of the Indian Census." *American Economic Journal: Economic Policy* 15 (2): 395–423. 10.1257/pol.20210129.
- Lobell, David B. 2007. "Changes in diurnal temperature range and national cereal yields." *Agricultural and Forest Meteorology* 145 (3–4): 229–238. 10.1016/j.agrformet.2007.05.002.
- Mall, Rajesh Kumar, Manisha Chaturvedi, Nidhi Singh, Rajeev Bhatla, Ravi Shankar Singh, Akhilesh Gupta, and Dev Niyogi. 2021. "Evidence of asymmetric change in diurnal temperature range in recent decades over different agro-climatic zones of India." *International Journal of Climatology* 41 (4): 2597–2610. 10.1002/joc.6978.

- Matsuyama, Kiminori. 1992. "Agricultural productivity, comparative advantage, and economic growth." *Journal of Economic Theory* 58 (2): 317–334. 10.1016/0022-0531(92)90057-0.
- Mundlak, Yair, Rita Butzer, and Donald F. Larson. 2012. "Heterogeneous Technology and Panel Data: The Case of the Agricultural Production Function." *Journal of Development Economics* 99 (1): 139–149. 10.1016/j.jdeveco.2011.10.006.
- Munshi, Kaivan and Mark Rosenzweig. 2016. "Networks and Misallocation: Insurance, Migration, and the Rural-Urban Wage Gap." *American Economic Review* 106 (1): 46–98. 10.1257/aer.20131365.
- Nath, Ishan. 2025. "Climate Change, the Food Problem, and the Challenge of Adaptation through Sectoral Reallocation." *Journal of Political Economy*. 10.1086/734725.
- Ortiz-Bobea, Ariel, Toby R. Ault, Carlos M. Carrillo, Robert G. Chambers, and David B. Lobell. 2021. "Anthropogenic Climate Change Has Slowed Global Agricultural Productivity Growth." *Nature Climate Change* 11 (4): 306–312. 10.1038/s41558-021-01000-1.
- Peng, Shaobing, Jianliang Huang, John E. Sheehy, Rebecca C. Laza, Romeo M. Visperas, Xuhua Zhong, Grace S. Centeno, Gurdev S. Khush, and Kenneth G. Cassman. 2004. "Rice yields decline with higher night temperature from global warming." *Proceedings of the National Academy of Sciences* 101 (27): 9971–9975. 10.1073/pnas.0403720101.
- Pham, Trinh. 2026. "Temperature Shocks, Worker Age, and Sectoral Reallocation into Informality: Evidence from Vietnam." Working paper, KDI School of Public Policy and Management.
- Ruggles, Steven, Lara Cleveland, Rodrigo Lovaton, Sula Sarkar, Matthew Sobek, Derek Burk, Dan Ehrlich, Quinn Heimann, and Jane Lee. 2024. "Integrated Public Use Microdata Series, International: Version 7.5 [dataset]." <https://doi.org/10.18128/D020.V7.5>.
- Saxena, Rimjhim. 2024. "Barriers within Borders: Structural Transformation and Climate Change in India." Discussion Papers in Economics Working Paper 24-05, University of Colorado Boulder, Department of Economics.
- Schlenker, Wolfram, W. Michael Hanemann, and Anthony C. 2006. "The Impact of Global Warming on U.S. Agriculture: An Econometric Analysis of Optimal Growing Conditions." *Rev. Econ. Stat.* 88 (1): 113–125. <https://ideas.repec.org/a/tpr/restat/v88y2006i1p113-125.html>.
- Schlenker, Wolfram and Michael J. Roberts. 2009. "Nonlinear temperature effects indicate severe damages to U.S. crop yields under climate change." *Proceedings of the National Academy of Sciences* 106 (37): 15594–15598. 10.1073/pnas.0906865106.
- Somanathan, Eswaran, Rohini Somanathan, Anant Sudarshan, and Meenu Tewari. 2021. "The Impact of Temperature on Productivity and Labor Supply: Evidence from Indian Manufacturing." *Journal of Political Economy*. 10.1086/713733.
- Stull, Roland. 2011. "Wet-Bulb Temperature from Relative Humidity and Air Temperature." *Journal of Applied Meteorology and Climatology* 50 (11): 2267–2269. 10.1175/JAMC-D-11-0143.1.
- Welch, Jarrod R., Jeffrey R. Vincent, Maximilian Auffhammer, Pura F. Moya, Achim Dobermann, and David Dawe. 2010. "Rice yields in tropical/subtropical Asia exhibit large but opposing sensitivities to minimum and maximum temperatures." *Proceedings of the National Academy of Sciences* 107 (33): 14562–14567. 10.1073/pnas.1001222107.
- Zhong, Ziqian, Bin He, Hans W. Chen, Deliang Chen, Tianjun Zhou, Wenjie Dong, Cunde Xiao, Shang-ping Xie, Xiangzhou Song, Lanlan Guo, et al. 2023. "Reversed asymmetric warming of

sub-diurnal temperature over land during recent decades." *Nature Communications* 14: 7189.
10.1038/s41467-023-43007-6.

Appendix: Summary Statistics and Supporting Exhibits

This appendix collects the exhibits a reader needs to reproduce the paper's figures and to check the model's predicted signs against the estimates. The full battery of robustness checks, the general-equilibrium model with all proofs, and the inference and measurement diagnostics are in the online appendix.

TABLE A1. Model comparative statics: predicted signs and their empirical counterparts

Outcome	Predicted sign		Empirical counterpart
	T_{min} (night)	T_{max} (day)	
Cultivated area	(−)	(−)	Equal contraction; $F = 0.01$ (Table 1)
Harvest price	(≈ 0)	(+)	Only daytime raises price; $F = 13.3$ (Table 1)
Field wage	(≈ 0)	(−)	Daytime wage loss (Table 1)
Cereal yield	($\approx 0, +$)	($\approx 0, -$)	Flat in dry-bulb; +0.15 wet-bulb night (online app.)
Ag. wage-labour share	(+)	(−)	Night raises, day lowers; $F = 24.9$ (Table 2)
Seasonal share	(−)	(+)	Opposite of wage labour; $F = 14.7$ (Table 2)
Cultivator share	(−)	(≈ 0)	Drawn off at night (Table 2)
Rural vs. urban amplitude	same sign, rural > urban		$F = 30.9$ vs. 1.9 (Figure 2)
Towns (non-agriculture)	(−)	(−)	Both margins depress; $F = 2.8$ (Figure 3)
Closed economy	signs reverse		Not observed $\Rightarrow$ open ($\chi < \chi^*$)

Notes: Signs are the framework's predictions in the empirically selected partial-open regime $0 < \chi < \chi^*$ under gross substitutes ($\sigma > 1$); the price asymmetry is the over-identifying sign restriction and the area equality is a calibration rather than a theorem. Only signs and the ordering of magnitudes are predicted, not levels. The full model, proofs, and the wet-bulb land-augmentation evidence are in the online appendix.

TABLE A2. Summary statistics: district and town, panel and long-difference samples

	Decadal panel (levels) 1981–2011		Long difference (Δ 2001–11 vs. 1981–91)	
	District	Town	District	Town
Cultivators (share)	0.323 (0.158)	0.080 (0.091)	-0.139 (0.080)	-0.058 (0.057)
Agricultural labourers (share)	0.180 (0.111)	0.088 (0.097)	-0.049 (0.057)	-0.053 (0.060)
Seasonal / marginal (share)	0.173 (0.111)	0.085 (0.079)	0.139 (0.086)	0.102 (0.065)
Non-agricultural (share)	0.324 (0.176)	0.745 (0.178)	0.048 (0.070)	0.010 (0.089)
Nighttime T_{min} (°C)	17.95 (2.39)	18.23 (2.22)	0.28 (0.24)	0.29 (0.24)
Daytime T_{max} (°C)	29.71 (1.95)	29.89 (1.63)	0.15 (0.19)	0.11 (0.20)
Rainfall (growing-season mean)	1.0 (0.5)	1.0 (0.5)	0.0 (0.1)	0.0 (0.1)
Units (districts / towns)	300	2,969	299	2,969
Observations	1,187	11,876	299	2,969

Notes: Means with standard deviations in parentheses, for the decadal panel (levels, 1981–2011) and the long-difference cross-section (Δ 2001–11 vs. 1981–91), each split into the district (total-population) and Census-town samples. Worker shares are of total workers in the unit; temperature and rainfall are growing-season ten-year moving averages. The town columns use the balanced panel of towns present in every round since 1981—the long-difference sample—so the panel-levels and long-difference town statistics describe the same 2,969 towns. The district panel counts one more unit than the long difference (300 vs. 299): one district is observed only at the endline and so enters the panel but cannot be differenced.

TABLE A3. District worker reallocation, long difference (1981–2011), by residence

	Share of Workers				Log Total Workers (5)
	Ag. Labour (1)	Cultivator (2)	Seasonal (3)	Non- Agriculture (4)	
<i>Panel A. Total population</i>					
ΔT_{min}	0.075 (3.58)	-0.024 (-0.79)	-0.074 (-2.74)	0.026 (0.94)	0.297 (1.24)
ΔT_{max}	-0.062 (-2.51)	-0.008 (-0.22)	0.049 (1.47)	0.020 (0.55)	0.041 (0.16)
F ($\Delta T_{min} = \Delta T_{max}$)	26.07	0.16	12.34	0.03	0.83
<i>p</i> -value	<0.001	0.687	<0.001	0.865	0.363
Observations	293	293	293	293	293
<i>Panel B. Rural population</i>					
ΔT_{min}	0.096 (4.12)	-0.035 (-1.11)	-0.064 (-2.25)	0.007 (0.37)	0.211 (0.91)
ΔT_{max}	-0.068 (-2.52)	0.011 (0.28)	0.056 (1.58)	-0.001 (-0.02)	0.025 (0.10)
F ($\Delta T_{min} = \Delta T_{max}$)	30.87	1.24	11.36	0.07	0.48
<i>p</i> -value	<0.001	0.266	<0.001	0.794	0.489
Observations	291	291	291	291	293
<i>Panel C. Urban population</i>					
ΔT_{min}	0.034 (2.64)	-0.013 (-0.89)	-0.028 (-2.09)	0.007 (0.25)	0.632 (1.75)
ΔT_{max}	0.006 (0.32)	0.015 (0.89)	0.006 (0.37)	-0.028 (-0.75)	0.305 (0.80)
F ($\Delta T_{min} = \Delta T_{max}$)	1.89	1.69	3.23	0.69	0.60
<i>p</i> -value	0.171	0.194	0.073	0.407	0.439
Observations	291	291	291	291	293
Rainfall Control	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y

Notes: Long-difference estimates of nighttime (ΔT_{min}) and daytime (ΔT_{max}) growing-season surface-air warming on district worker shares, decomposed into the total, rural, and urban populations of each district (the within-district residence split). These are the coefficients behind Figure 2: the agricultural wage-labour column of Panels B and C is plotted there. Point estimates with $|t| > 2$ in bold over district-clustered *t*-statistics; the *F*-statistic and its *p*-value test $\Delta T_{min} = \Delta T_{max}$. All specifications include state fixed effects and a precipitation control.

TABLE A4. Town worker reallocation, long difference (1981–2011)

	Share of Workers				Log Total Workers
	Ag. Labour (1)	Cultivator (2)	Seasonal (3)	Non-Agriculture (4)	
ΔT_{min}	0.063 (5.46)	-0.009 (-0.86)	-0.016 (-1.35)	-0.038 (-2.16)	0.272 (1.61)
ΔT_{max}	0.041 (2.59)	0.022 (1.49)	0.017 (1.15)	-0.078 (-3.47)	0.241 (1.41)
Observations	2,969	2,969	2,969	2,969	2,969
F-stat ($\Delta T_{min} = \Delta T_{max}$)	1.61	3.38	4.06	2.78	0.11
<i>p</i> -value	0.205	0.067	0.044	0.096	0.738
Rainfall Control	Y	Y	Y	Y	Y
FE / Trend	Y	Y	Y	Y	Y

Notes: Census-town worker reallocation under nighttime (ΔT_{min}) and daytime (ΔT_{max}) growing-season surface-air warming, across the balanced panel of Census towns—the coefficients behind Figure 3. The four mutually exclusive worker categories (agricultural wage labour, cultivators, seasonal, non-agricultural) sum to one, so the share coefficients sum to approximately zero; column 5, set apart, reports the long-run change in log total town workers. Point estimates with $|t| > 2$ in bold over district-clustered *t*-statistics; the *F*-statistic and its *p*-value test $\Delta T_{min} = \Delta T_{max}$. State fixed effects and a precipitation control throughout.